

Project Report: Stone Paper Scissor Game

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1. Introduction

The **Stone Paper Scissor Game** is a console-based application developed in Python. It is a digital version of the classic hand game usually played between two people. In this version, the player competes against an automated computer opponent that makes randomized selections.

2. Project Objectives

- To implement basic Python programming concepts (variables, data types, and input/output).
- To demonstrate the use of conditional logic (**if-elif-else**) for decision-making.
- To utilize the **random** library for simulating AI behavior.
- To create a mapping system using Dictionaries for user-friendly inputs.

3. System Requirements

- **Language:** Python 3.x
- **Libraries:** **random** (Built-in)
- **Interface:** Command Line Interface (CLI)

4. Design & Logic

The game operates on a numerical mapping system to simplify the comparison logic:

Choice	Numeric Value
Stone	0
Paper	1
Scissor	-1

Logic Flow:

1. **User Input:** The user provides a shorthand string (st, p, sc).
2. **Computer Selection:** The computer randomly selects an integer from the set {0, 1, -1}.
3. **Mapping:** The program converts the user's string into the corresponding integer.
4. **Comparison:** * If User == Computer, it is a Draw.
 - Specific win/loss conditions are checked using conditional branches to determine the winner.
5. **Output:** The final result is printed to the console with a message.

5. Implementation (Source Code)

```
Python
import random

# Core logic and mapping
options = {"st":0, "p":1, "sc":-1}
c_choice = {0:"Stone", 1:"Paper", -1:"Scissor"}

computer = random.choice([0, 1, -1])
youchose = input("Enter your choice (st/p/sc): ")
you = options[youchose]

print(f"Your choice is: {c_choice[you]}")
print(f"Computer choose: {c_choice[computer]}")

# Evaluation
if(computer == you):
    print("Its a draw!")
else:
    # Win Conditions
    if (computer == 0 and you == 1) or \
        (computer == 1 and you == -1) or \
        (computer == -1 and you == 0):
        print("You won! \nCongratulations")
    # Loss Conditions
    else:
        print("You lose! \nBetter Luck Next Time.")
```

6. Testing & Results

The application was tested against various scenarios:

- **Scenario A:** User chooses 'st', Computer chooses 'sc' -> **Result: You won.**
- **Scenario B:** User chooses 'p', Computer chooses 'p' -> **Result: Its a draw.**
- **Scenario C:** User chooses 'sc', Computer chooses 'st' -> **Result: You lose.**

7. Conclusion

The project successfully achieves the goal of a functional, interactive game. It demonstrates how real-world logic can be translated into code using Python.

8. Future Enhancements

- **Score Tracker:** Adding a variable to keep track of wins over multiple rounds.
- **Multiplayer Mode:** Allowing two users to play against each other.
- **GUI:** Developing a graphical user interface using **Tkinter** or **Pygame**.